an opponent raised his hackles. But now, liars might start to evolve. Individuals who really had no intention of going on for a long time would raise their hackles on every occasion, and reap the benefits of easy and quick victory. So liar genes would spread. When liars became the majority, selection would now favour individuals who called their bluff. Therefore liars would decrease in numbers again. In the war of attrition, telling lies is no more evolutionarily stable than telling the truth. The poker face is evolutionarily stable. Surrender, when it finally comes, will be sudden and unpredictable.

So far we have considered only what Maynard Smith calls ‘symmetric’ contests. This means we have assumed that the contestants are identical in all respects except their fighting strategy. Hawks and doves are assumed to be equally strong, to be equally well endowed with weapons and with armour, and to have an equal amount to gain from winning. This is a convenient assumption to make for a model, but it is not very realistic. Parker and Maynard Smith went on to consider asymmetric contests. For example, if individuals vary in size and fighting ability, and each individual is capable of gauging a rival’s size in comparison to his own, does this affect the ESS that emerges? It most certainly does.

There seem to be three main sorts of asymmetry. The first we have just met: individuals may differ in their size or fighting equipment. Secondly, individuals may differ in how much they have to gain from winning. For instance an old male, who has not long to live anyway, might have less to lose if he is injured than a young male with the bulk of his reproductive life ahead of him.

Thirdly, it is a strange consequence of the theory that a purely arbitrary, apparently irrelevant, asymmetry can give rise to an ESS, since it can be used to settle contests quickly. For instance it will usually be the case that one contestant happens to arrive at the location of the contest earlier than the other. Call them ‘resident’ and ‘intruder’ respectively. For the sake of argument, I am assuming that there is no general advantage attached to being a resident or an intruder. As we shall see, there are practical reasons why this assumption may not be true, but that is not the point. The point is that even if there were no general reason to suppose that residents have an advantage over intruders, an ESS depending on the asymmetry itself would be likely to evolve. A simple